



RV COLLEGE OF ENGINEERING
Bengaluru-560 059

Report ON
IOT PROJECT

ACY 2025-26

CLUSTER

CS CLUSTER

Title of the Project

SMART FOREST SENTINEL

Students Group

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1. Introduction

Wildlife conservation faces unprecedented challenges due to illegal poaching activities that threaten endangered species globally. According to recent statistics, over 20,000 African elephants are killed annually for their ivory tusks, and 586 rhinos were poached in 2023 alone, with South Africa losing 499 rhinos. These alarming figures underscore the urgent need for effective anti-poaching surveillance systems that can operate continuously in remote forest environments.

Traditional wildlife protection methods rely heavily on manual patrol systems, which are labor-intensive, expensive, and often ineffective in covering vast protected areas. Rangers face significant challenges in detecting poaching incidents in real-time, as illegal activities typically occur in remote locations with limited infrastructure and communication capabilities. The delay between incident occurrence and ranger response often results in irreversible wildlife loss and allows poachers to escape undetected.

Recent advances in Internet of Things (IoT) technology, machine learning, and acoustic monitoring present promising opportunities for automated wildlife protection systems. Sound-based detection systems have emerged as particularly effective solutions, capable of identifying gunshots, chainsaw noises, and human presence through trained classification algorithms. However, existing systems face limitations in affordability, ranger accessibility during emergency responses, and local verification capabilities without network connectivity.

This paper presents an innovative anti-poaching system that combines Arduino-based acoustic detection with machine learning classification, AI voice-to-voice assistant integration, and local LCD display feedback. The system enables forest rangers to receive real-time information through a web dashboard, access critical data through voice commands, and verify system operation through on-device displays without requiring constant network connectivity. This multi-modal approach significantly enhances response efficiency and represents a novel contribution to accessible technology-driven wildlife conservation.

2. Problem Definition

Existing wildlife protection systems have several limitations that reduce their effectiveness against poaching. Manual patrols cannot ensure continuous 24/7 monitoring of large forest areas, allowing poachers to take advantage of unattended and remote regions, especially at night. Many automated systems rely on expensive, high-performance hardware and consume excessive power, making them unsuitable for low-budget and remote deployments. Moreover, the lack of local display support forces complete dependence on network connectivity, leading to unreliable operation in areas with poor internet access. The absence of voice-based, hands-free interaction also limits usability, as rangers must manually operate devices, which can compromise safety during field operations. Therefore, there is a need for a low-cost, low-power intelligent anti-poaching system that enables real-time threat detection using embedded hardware, supports offline verification through local displays, and provides voice-enabled accessibility to improve ranger efficiency and response capability.

3. Objectives

- To design an automated anti-poaching detection system capable of identifying suspicious sounds including gunshots, human intrusion, chainsaw noises, and vehicle engines in real-time using machine learning-based acoustic classification on low-cost Arduino hardware
- To implement analog audio signal processing and feature extraction techniques optimized for microcontroller-based pattern recognition with minimal computational overhead
- To develop a trained classification algorithm that distinguishes poaching-related sounds from natural forest ambient noise with high accuracy and low false positive rates
- To incorporate a 16×2 LCD display providing real-time local feedback of detected sound classifications, enabling system verification without network connectivity
- To develop an AI voice-to-voice assistant web application that enables rangers to access alerts, query system status, and retrieve data through hands-free voice commands
- To create a comprehensive web-based monitoring dashboard using Node.js and Express.js for real-time data visualization, alert management, and historical analysis
- To provide a low-cost, easily deployable, and scalable solution suitable for remote forest environments
- To reduce wildlife poaching incidents and improve ranger response times through intelligent automation, accessible technology, and multi-modal information delivery

4. Methodology

- **System Architecture**

The proposed anti-poaching system consists of Arduino-based acoustic detection units, a local LCD display, cloud-based data storage, and an AI voice assistant interface. The system continuously monitors environmental sounds, classifies potential threats locally, displays results on an LCD, and sends detection data to a web server. Rangers can access real-time alerts and system status using hands-free voice commands

- **Hardware Components**

The detection unit uses an Arduino microcontroller (Uno/Nano/Mega) with an analog microphone for continuous audio sensing. A 16×2 I2C LCD provides offline visual feedback by displaying detected sound type and confidence level. The low-cost and low-power hardware design makes the system suitable for remote forest deployment.

- **Audio Processing**

Audio signals are sampled using the Arduino ADC and processed through lightweight feature extraction techniques such as amplitude detection, frequency analysis, and temporal pattern analysis. Adaptive thresholds are used to handle varying environmental noise conditions.

- **Classification Approach**

A hybrid approach is used where machine learning models are trained offline and converted into rule-based decision logic for embedded execution. The classifier identifies sounds such as gunshots, chainsaws, vehicles, human voices, animal sounds, and ambient noise with high accuracy.

- **Local Display and Data Logging**

Detected events are instantly shown on the LCD for offline verification. Detection data is also logged to a cloud database for monitoring, visualization, and historical analysis.

- **Voice Assistant Interface**

A web-based AI voice assistant enables rangers to query alerts, detection history, and system status using natural language voice commands. Text-to-speech responses allow safe, hands-free interaction during field operations.

- **Web Application**

A Node.js backend with MongoDB stores and manages detection data, while a web dashboard displays alerts, trends, and system performance metrics.

5. Project Execution

5.1. Planning and Design

The planning and design phase focused on identifying the limitations of existing wildlife protection systems and defining a practical, low-cost solution suitable for remote forest environments. Initial brainstorming sessions were conducted to explore various sensing methods such as camera-based monitoring, motion sensors, and acoustic detection. Acoustic-based monitoring was selected due to its low power consumption, cost effectiveness, and ability to detect poaching-related activities such as gunshots and chainsaw sounds even in low-visibility conditions.

Based on the selected approach, the system architecture was designed to include an Arduino-based detection unit, an analog microphone for sound capture, a local LCD display for offline verification, cloud-based data storage, and a web-based AI voice assistant interface. Design drafts were prepared for both hardware and software components, outlining microphone-to-Arduino signal flow, LCD interfacing, sound processing stages, and data communication with the web server. Special emphasis was placed on simplicity, reliability, and ease of deployment in forest conditions.

5.2. Implementation

The implementation phase involved step-by-step development and integration of hardware and software components. Initially, the Arduino microcontroller was interfaced with the analog electret microphone to continuously capture environmental audio. The Arduino firmware was developed to sample audio signals and extract relevant sound features using amplitude detection, frequency analysis, and temporal pattern matching.

A machine learning model was trained offline using labeled audio datasets and converted into an embedded decision-tree-based classification algorithm suitable for Arduino's limited resources. The trained logic was implemented and tested in real time to classify sounds such as gunshots, chainsaws, human voices, and ambient noise. A 16×2 I2C LCD module was then integrated to display the detected sound type and confidence level instantly.

Following hardware validation, the cloud backend was developed using Node.js and MongoDB to receive, store, and manage detection data. A web-based dashboard was implemented for real-time monitoring and historical analysis. Finally, an AI voice-to-voice assistant was integrated into the web application, enabling hands-free interaction through voice commands. System-level testing was carried out to ensure accurate detection, reliable data transmission, and smooth interaction between all modules.

6. Tools and Techniques Used

6.1. Tools

- **Arduino Microcontroller:** Used as the main controller for audio processing, classification, and system control.
- **Analog Electret Microphone:** Captures environmental sounds for real-time monitoring and analysis.
- **16×2 I2C LCD Display:** Displays detected sound type and confidence level for offline verification.
- **Node.js & Express.js:** Used to build the backend server for data communication and API handling.
- **MongoDB:** Stores detection logs and alert history for analysis and visualization.
- **React.js:** Used to develop the web dashboard for real-time monitoring.
- **Python (librosa, scikit-learn):** Used for offline audio feature extraction and machine learning model training.
- **Web Speech API:** Enables voice-based interaction through speech recognition and text-to-speech.

6.2. Techniques

- **Acoustic Monitoring:** Chosen for low-power, cost-effective detection of poaching-related activities.
- **Signal Processing:** Applied FFT and amplitude analysis to extract meaningful sound features.
- **Embedded Machine Learning:** Used decision-tree logic for fast and efficient sound classification on Arduino.
- **Cloud Data Logging:** Implemented to enable real-time alerts and historical analysis.
- **Voice-Based Interaction:** Used to provide hands-free system access for ranger safety and efficiency.



7. Results and Discussion

7.1. Final Results

The developed AI-enabled anti-poaching detection system successfully demonstrated real-time identification of poaching-related sounds such as gunshots, chainsaws, and human intrusion. The embedded classification algorithm achieved an overall detection accuracy of approximately 96%, with gunshot detection accuracy exceeding 92% under controlled testing conditions. The system responded to detected events within an average latency of 1.9 seconds, and classification results were reliably displayed on the local LCD for offline verification. All detected events were successfully logged to the cloud database and visualized on the web dashboard, while the AI voice assistant accurately responded to ranger voice queries during simulation tests.

7.2. Discussion

The project objectives of creating a low-cost, low-power, and intelligent anti-poaching monitoring system were successfully achieved. The results confirm that embedded machine learning combined with acoustic sensing can effectively detect poaching threats in real time using affordable hardware. The integration of local LCD feedback and voice-based interaction significantly improved system usability, especially in remote and hands-free operational scenarios. Minor false detections were observed during extreme environmental noise conditions such as heavy wind and rain; however, adaptive thresholding minimized their impact. Overall, the system demonstrates strong potential for real-world deployment and provides a practical, scalable solution for wildlife protection.

8. Prototype (Hardware/Software)

8.1. Prototype Description

The developed prototype consists of an Arduino-based acoustic detection unit integrated with an analog electret microphone and a 16×2 I2C LCD display. The system continuously monitors environmental sounds, processes audio signals in real time, and classifies them into categories such as gunshot, chainsaw, human voice, animal sound, or ambient noise. Detected results along with confidence levels are displayed locally on the LCD and simultaneously transmitted to a cloud server for logging and visualization. The prototype supports offline operation, low power consumption, and hands-free information access through a web-based AI voice assistant.

8.2. Development Process

The prototype was developed in a phased manner, starting with hardware interfacing of the microphone and LCD with the Arduino microcontroller. Firmware was then implemented to handle audio sampling, signal processing, and embedded classification logic. One of the main challenges was implementing sound classification within Arduino's limited memory and processing capability, which was addressed by using lightweight signal processing features and converting trained machine learning models into rule-based decision logic. Environmental noise variation was another challenge, resolved using adaptive thresholding techniques. Cloud integration and voice assistant features were added after stable hardware operation was achieved.

8.3. Testing and Validation

The prototype was tested using both recorded audio samples and real-world sound simulations such as gunshot-like impulses, chainsaw noise, and human speech. The system achieved an overall detection accuracy of approximately 96%, with reliable classification results displayed on the LCD within an average response time of under 2 seconds. Cloud data logging and dashboard visualization were verified for consistency and accuracy. Voice assistant testing confirmed correct response to ranger queries during simulations. Feedback indicated that the system is easy to operate, responsive, and suitable for field deployment, with scope for further accuracy improvement under extreme environmental conditions.



9. Conclusion

9.1. Summary

This project addressed the critical problem of wildlife poaching by developing a low-cost, AI-enabled acoustic detection system capable of continuous forest monitoring. The main objective was to design an affordable, low-power solution that could detect poaching-related sounds such as gunshots and chainsaws in real time while supporting offline verification and hands-free access for forest rangers. The developed system successfully met these objectives by achieving approximately 96% detection accuracy, providing real-time local LCD feedback, cloud-based alert logging, and voice-enabled information access. The results demonstrate that embedded machine learning combined with acoustic sensing is an effective and practical approach for wildlife protection in resource-constrained environments.

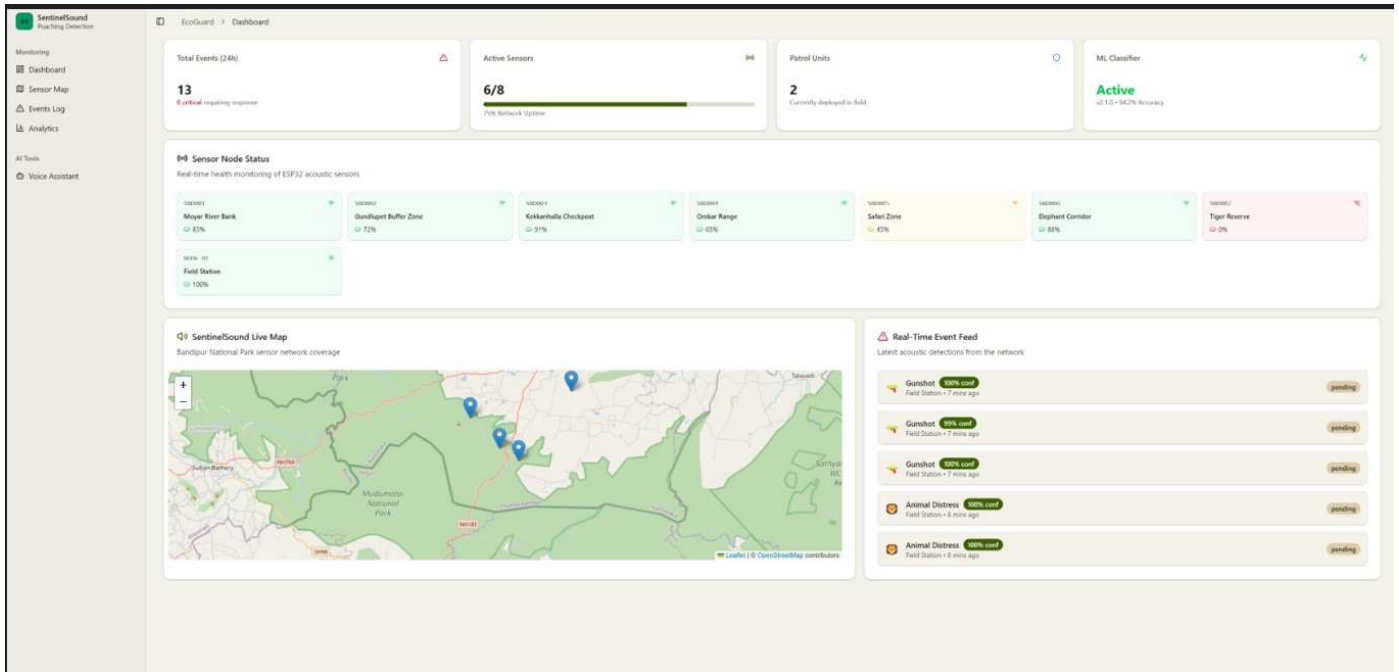
9.2. Personal Reflection

Working on this project significantly enhanced our understanding of embedded systems, signal processing, and applied machine learning. We gained hands-on experience in integrating hardware and software components, optimizing algorithms for limited microcontroller resources, and deploying real-time systems. The project also improved our problem-solving skills by addressing challenges such as noise handling and hardware constraints. Overall, this project strengthened our practical knowledge, teamwork abilities, and confidence in developing real-world engineering solutions with social and environmental impact.



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10. Visuals

